Bangladesh University of Engineering and Technology
Department of Electrical and Electronic Engineering
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Brain Tumor Detection & Classification

Submitted to:

Dr. Maruf Ahmed

Assistant Professor, Department of EEE

Bangladesh University of Engineering & Technology

Shafin Shadman Ahmed

Lecturer, Department of EEE

Bangladesh University of Engineering & Technology

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY

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Digital Signal Processing I Laboratory

Final Project Report

Section: C2 Group: 04

Brain Tumor Detection & Classification

Course Instructors:

Dr. Maruf Ahmed, Assistant Professor, EEE, BUET

Shafin Shadman Ahmed, Lecturer, EEE, BUET

Signature of Instructor: _____

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Signature: _____ **Full Name: Md. Nagib Mahfuz** **Student ID: 2006166**	**Signature:** _____ **Full Name: Muntafa Mubarrat Mahi** **Student ID: 2006172**
Signature: _____ **Full Name: Mahmud Al Hadi Nafis** **Student ID: 2006178**	**Signature:** _____ **Full Name: Mushfikur Rahman Navin** **Student ID: 2006190**

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1.Abstract

Brain tumor classification is critical for evaluating tumors and making therapy decisions. There are several imaging modalities used to identify brain tumors. MRI is widely utilized for its high picture quality and lack of ionizing radiation. Deep learning (DL), a subfield of machine learning, has lately demonstrated exceptional performance in classification and segmentation tasks. Our project proposes a convolutional neural network(CNN)-based deep learning model for classifying brain tumor types utilizing one publicly accessible dataset. This classifies tumors into three categories: meningioma, glioma, and pituitary tumor. The dataset includes 233 patients MRI image of three kinds with a total of 3064. The suggested network topology had the highest overall accuracy 87.9%) in our trial. The result shows that the model can effectively classify brain tumors.

2.Introduction

A brain tumor is an abnormal growth of cells in the brain. These tumors can be either benign (non-cancerous) or malignant (cancerous), and they can originate from brain cells or from cells in other parts of the body that have spread to the brain. Numerous imaging techniques can be used to detect and classify brain tumors. Meningiomas, gliomas, and pituitary tumors each have their own challenges when it comes to diagnosis and treatment However, MRI is one of the most common non-invasive techniques. MRI popularity comes from the fact of using no ionizing radiation during the scan as well as its superior soft-tissue resolution plus the ability to acquire different images using various imaging parameters or by employing contrast-enhanced agents. Detecting and categorizing brain cancers, such as meningiomas, gliomas, and pituitary tumors, has progressed because to technologies like as MATLAB and sophisticated algorithms known as Convolutional Neural Networks. These brain tumors are difficult to detect and treating them necessitates understanding their particular features.

Deep Learning (DL) is a subdivision of ML that is based on learning data representations and hierarchical feature learning. DL algorithms utilize arrange of numerous layers of nonlinear processing identities for feature extraction. The output of each sequential layer is the input of the next one, and that helps in data abstraction as we go deep within the network . Convolutional Neural Network (CNN) is a class of DL and commonly used in analyzing visual imagery and designed to require minimal preprocessing .It is inspired by biological processes in human brain and utilized to handle data that come in multiple arrays.

CNNs have the key benefits of feature learning and limitless accuracy over standard machine learning and vanilla neural networks, which may be obtained by increasing training samples, resulting in a more robust and accurate model. Convolutional filters in CNN architecture operate as feature extractors, and as we go deeper, we extract increasingly complicated features (spatial and structural information). Feature extraction occurs by convolving tiny filters with input patterns, followed by selecting the most distinctive features and then beginning to train the classification networks.

3 Design:

3.1 Problem Formulation (PO(b))

3.1.1 Identification of Scope

The scope of our project is:

- Early Detection: Timely identification of brain tumors for prompt treatment.
- Accuracy and Efficiency: High accuracy in diagnosis, optimizing resource use.
- Personalized Treatment: Tailored treatment plans for better outcomes.
- Cost Reduction: Lower healthcare costs through early intervention.
- Improved Patient Care: Enhanced diagnostic capabilities for better care.
- Professional Development: Opportunities for healthcare innovation.
- Time efficiency: Less Computational Time

3.1.2 Literature Review

The literature review of this project consists of the research papers, articles, and textbooks that were consulted regarding brain tumor detection and image processing and their application. Previous studies have explored various aspects of Brain Tumor detection from MRI image projects, focusing on image processing, early detection, accuracy and efficiency, cost reduction, safety considerations, and educational objectives. This literature review forms the basis for our project, guiding our approach and contributing to our understanding of brain tumor detection from MRI images.

3.1.3 Formulation of Problem

In our project, given a dataset of MRI brain images, our objective is to develop a deep-learning CNN model capable of accurately detecting and classifying brain tumors. The model should address challenges such as data scarcity, variability in image quality, interpretability, computational constraints, and class imbalance. Ethical considerations regarding patient privacy and responsible data use will be adhered to throughout the project.

- Data Availability: The limited availability of labeled MRI datasets with diverse tumor types and sizes poses a challenge for training robust models.
- Data Quality: Variability in MRI image quality due to factors like noise, artifacts, and scanner inconsistencies may affect model performance.
- Computational Resources: Training deep learning models requires significant computational resources, which may not be readily available in resource-constrained environments.
- Ethical Considerations: Ensuring patient privacy, data security, and ethical use of medical data throughout the project is the most important.

3.2 Design Method (PO(a))

The project aims to design a deep learning Convolutional Neural Network (CNN) model for the detection and classification of brain tumors from MRI images. For this purpose, at first, we collected data from the figshare.com website which was collected from Nanfang Hospital and General Hospital, Tianjing Medical University, China from 2005 to 2010. We took a total 3064 images of 233 patients with three types of brain tumors that are meningioma, glioma, and pituitary tumor.

After extracting and loading the image levels, we divided them into three distinct sections: training, testing, and validation. The main challenge of our project was designing the CNN model, where careful consideration was given to factors such as architecture selection, hyperparameter tuning, and addressing issues related to data scarcity and variability. Once the final model was designed, we conducted performance computation to evaluate its effectiveness. The entire design model, along with a flowchart, is presented below:

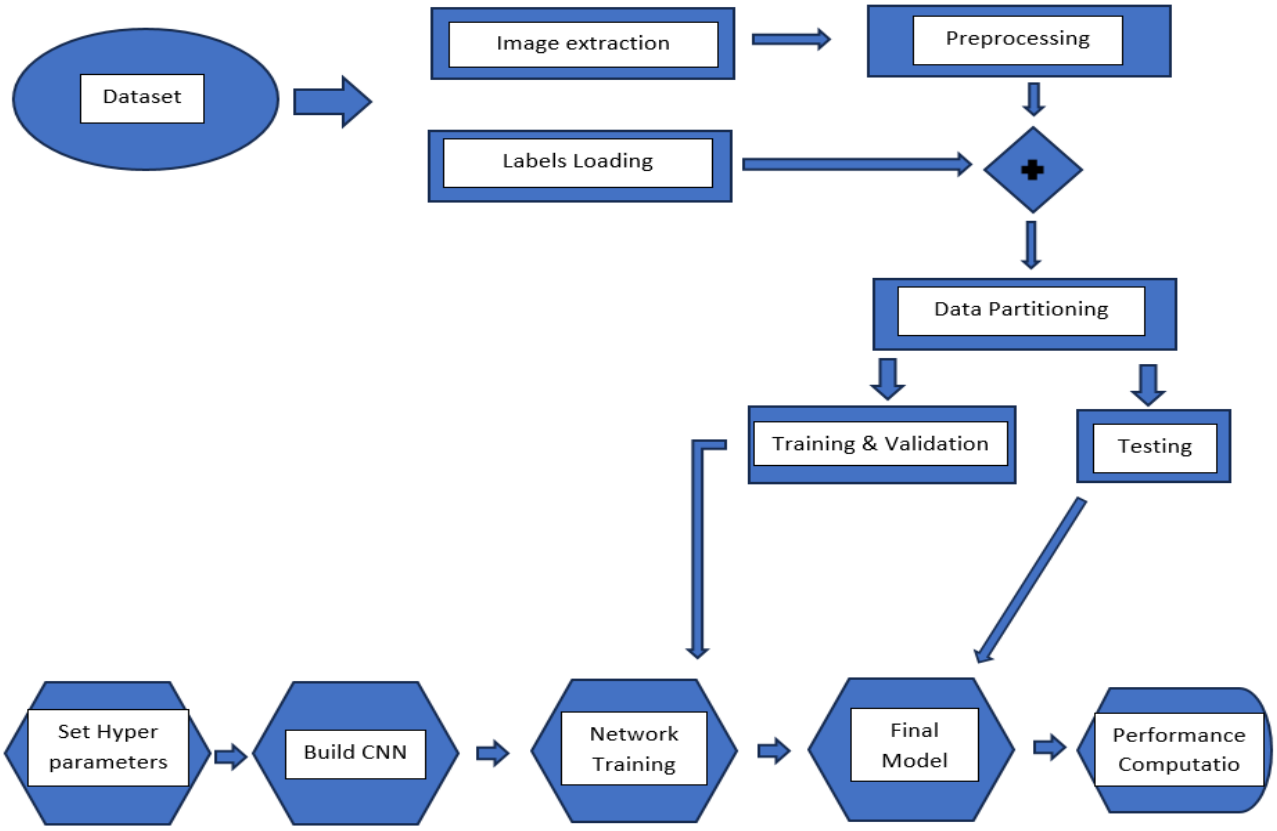


Fig: Flowchart of Brain Tumor Detection using CNN model

4.Implementation

A. Dataset

We utilized a dataset in this project. It was obtained from General Hospital, Tianjing Medical University, China, between 2005 and 2010, and then released online in various editions beginning in 2015 and ending in 2017. The collection contains T1-weighted contrast-enhanced pictures from 233 individuals with three kinds of brain tumors: meningioma, glioma, and pituitary tumor.
Brain tumors can vary in shape, location, and size depending on their nature. The dataset has three separate perspectives: axial, coronal, and sagittal views.

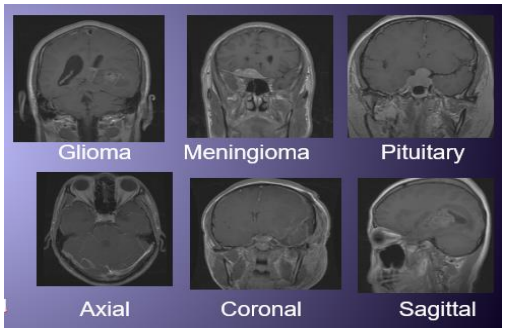


Fig: Sample image of MRI (from dataset)

B. Pre-processing Stage

Preprocessing is done on the photos before they are fed into the suggested framework. First, the original image must be reduced in size from 512 X 512 X 1 pixels into 128 X 128 X 1 pixels in an effort to reduce dimensionality, computations, and aid the network in demonstrating improved performance in shorter computation times.

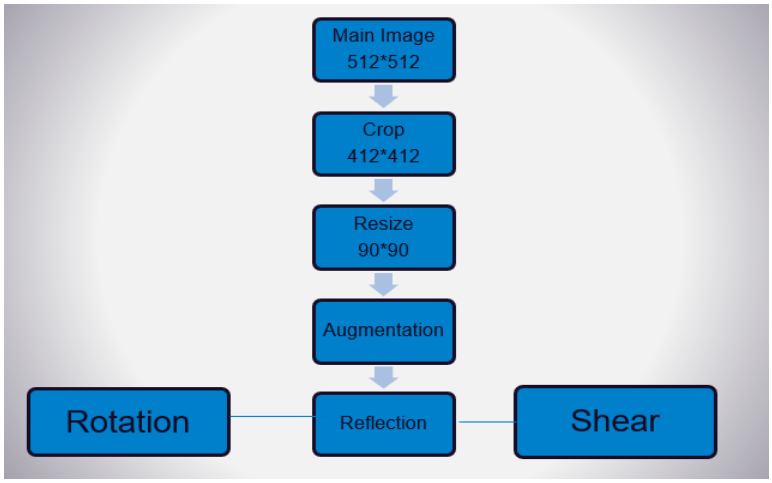


Fig : Preprocessing stages

To ensure that the system continues to learn on unsorted data and avoid concentrating on a small portion of the total dataset, data is then shuffled before being split. With their distinct target labels, the data is separated into three sections: training, validation, and test sets (68% for training and 12 % for system validation and 20% for testing). Finally, we augment the images of study I so that the system can identify them as new ones, and that is usually used to avoid overfitting and increase model robustness.

Our images were augmented to increase the number of images, so that we have sufficient number of datasets for training .In augmentation ,we made changes like flipping around the x-axis, right/left mirroring, shearing and image rotation . By doing this augmentation process, we have increased the original 3064 images by a factor of 5, so the original dataset became 15,320 images.

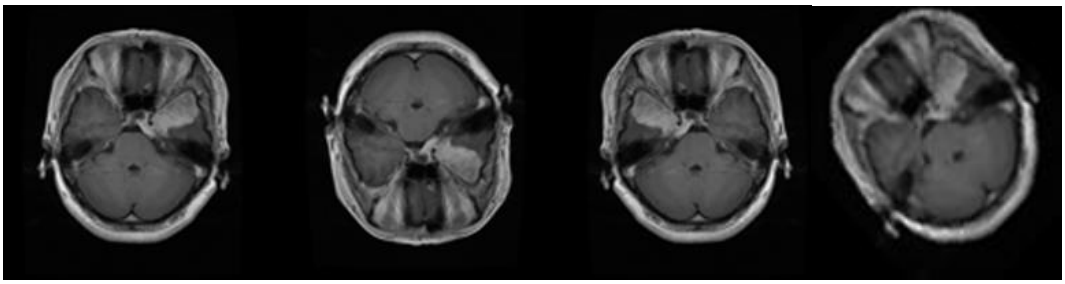


Fig : original, up-down flipping, right/left mirroring, rotating by 30 degrees

C. CNN model

The CNN model consists of 16 layers, with the input layer holding the enhanced images from the prior pre-processing stage and moving through the convolution layers with their corresponding activation functions. Number of slices used in features selection and downsampling (convolution, Rectified Linear Unit (ReLU), normalization, and pooling layers). A dropout layer is employed to prevent overfitting, and is followed by a fully connected layer, a softmax layer, and a layer that predicts the output before a classification layer generates the predicted class.

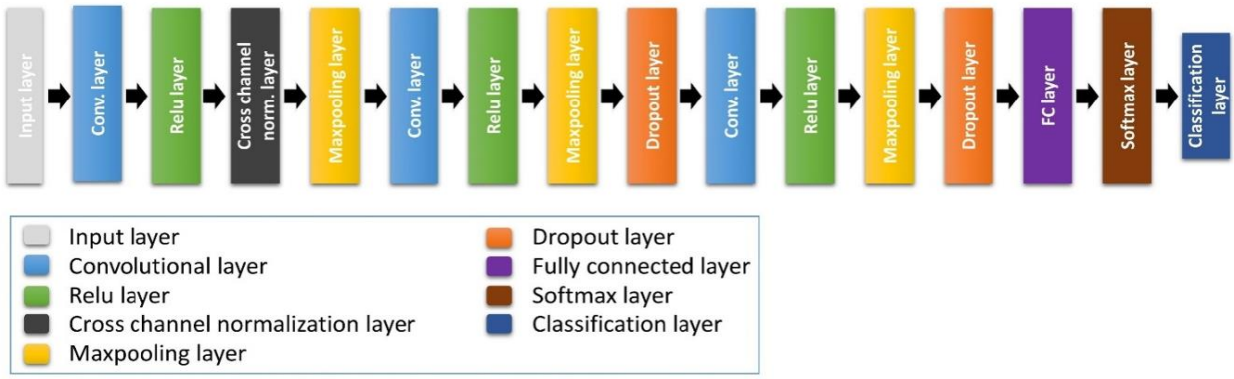


Fig . The proposed CNN architecture

The description of each layer is as follows; First, the input layer is used to confirm the size of input images and applying a data normalization . In the proposed architecture, three convolutional layers are included. A 2D convolutional layer applies sliding K convolutional filters (kernels) of size (M X N) over the input images by moving the filters along the input and compute the dot product of the weights (kernels weights) and the input.

The filters slide over the image with vertical and horizontal steps called stride (S). Padding (P) of the original images may happen before sliding the filters in order to maintain information at the edges. These kernels are used as features identifiers; such that kernels in the early layers detect only low-level features like (edges, lines and blobs), while advanced ones are used to detect more and more complex features.

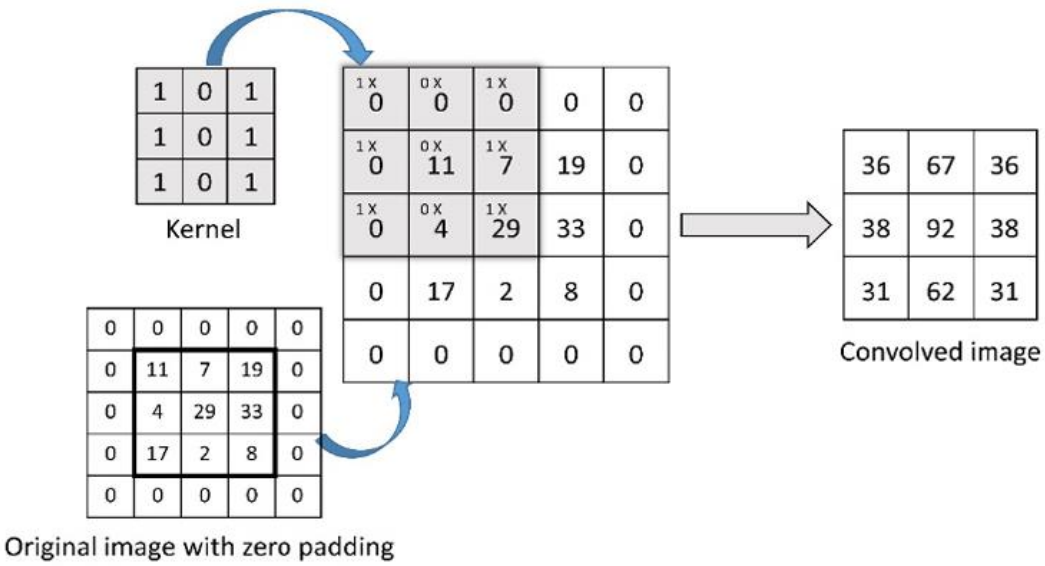


Figure :Convolutional layer example

A non-saturated activation function called ReLU follows each convolutional layer. Its primary purpose is to significantly reduce training time when compared to other activation functions. The ReLU model is expressed as a function of x in the following equation, where the output equals the input for positive values of x and equals 0 for other values. The input layer is normalized using a cross-channel normalization layer, which scales and modifies the associated activations. It creates a local response normalization layer using a channel-wise window of a specific size, which is selected at random. Normalization has applications in network training acceleration and backpropagation.

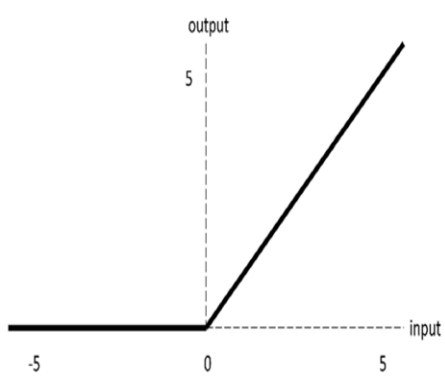


Fig : ReLU activation function

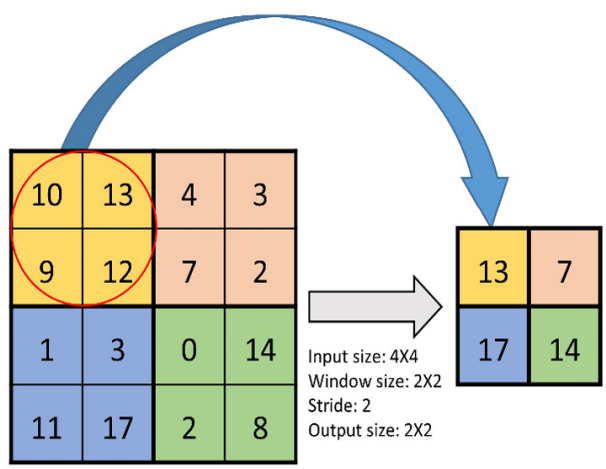


Figure : Example of a max-pooling layer

As for maximum Pooling layer, it is a way of down sampling used to achieve spatial invariance by splitting the whole image into small rectangles (2 X 2 in the proposed structure) that are moving over the image with a determined step (2 X 2) and then consider only the maximum value of the four elements. The pooling layer is used to reduce numbers of parameters and

consequently computations in the network.

One of the most common methods to reduce overfitting is to use a dropout layer (an example is shown in figure). In this layer, some activations (nodes) are dropped out randomly which significantly helps also in speeding up the training phase.

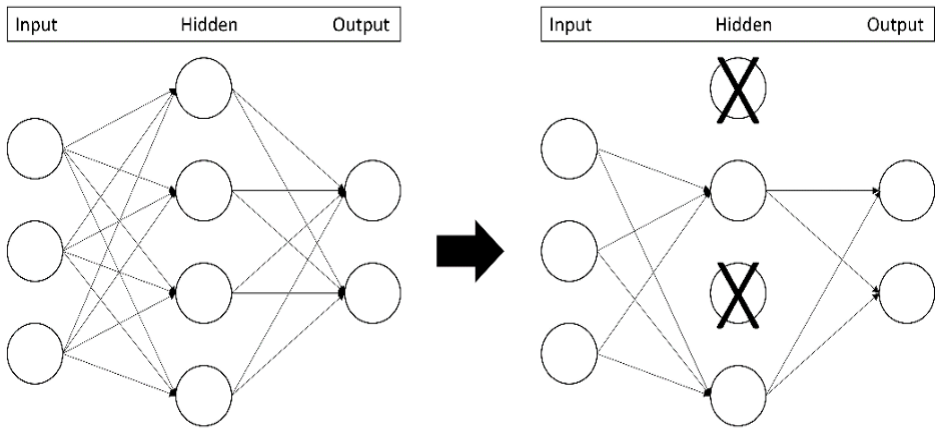


Figure **:** Example of dropout layer

Finally, we have used three advanced layers; Fully Connected layer (FC), softmax layer and classification layer. The former one is used to connect every neuron in a layer to every neuron in another one (following and preceding); the output of this layer is three classes. Then, the FC layer is followed by a softmax layer that is also called (normalized exponential function). Softmax layer is used to squash all the predicted classes between 0 and 1, and the total sum of these values is equal to 1 (100%). The output of this layer can be calculated as follows:

$$y(z)_j = \frac{e^{z_j}}{\sum_{k=1}^{k} e^{z_k}}$$

The probability of any class (j) can be calculated over (k) different classes as a function y (z) and their total summations are equal to as shown in figure. Finally, we use a classification layer which is based on cross-entropy loss to estimate the classification loss and provides the final predicted categorical label for each input image. Loss can be estimated from equation, where p is the target labels vector, and q (x) is the output vector from the softmax layer.

$$H(p, q) = -\sum_{x} (p(x) * log(q(x)))$$

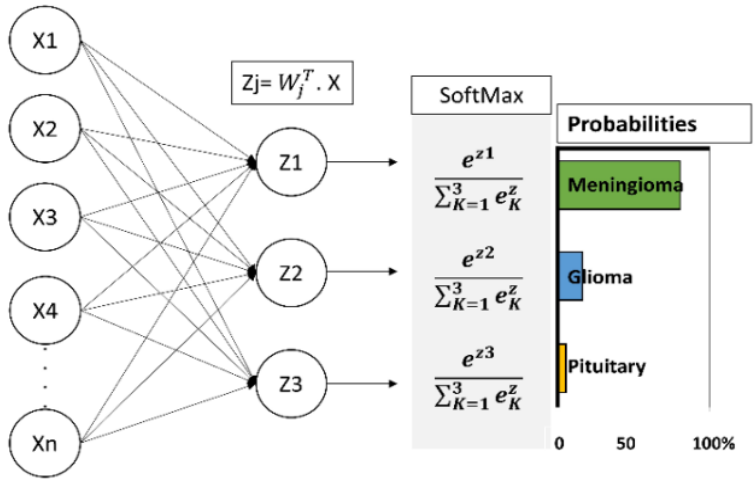


Figure: Example of Soft-max layer.

D. Regularization techniques and Optimization Algorithm

Regularization means fitting a solving function well during training while preventing system overfitting. Many techniques have been used to avoid overfitting during preprocessing and training phases. First, data augmentation is used to avoid overfitting by making a geometric and color distortion on the original images. Then, different network architectures have been tested to avert network complexity. Then, dropout layers have been used to remove hid den units weights stochastically. L2 regularization is also used to add a penalty to the cost function and introduce weights decay as shown in the following equation.

$$Cost\ function' = Cost\ function\ (Loss) + \lambda \sum_{i=1}^{k} w_i^2$$

where is the regularization parameter (hyper-parameter that is defined), w is the corresponding weight(s) for i=1……k. Finally early stop technique has been used in some trials which monitor the training and validation performance and stops the training process before full epochs training completion in case of system stability or before starting to overfit.

Optimization is used mainly to update network parameters and minimize loss function to reach the global minimum in the ideal case by taking small steps to the negative gradient direction (convergence). We have found that the stochastic gradient descent with momentum is the optimal optimizer for the proposed structure.

5 Design Analysis and Evaluation

5.1 Novelty

The project implements a brain tumor detection program using MRI images. The project does not take tumor regions as input unlike some previous approaches in this regard. In this regard, the primary novelty of the project is the fully automated method of detection. In addition, preprocessing required for the project includes cropping, resize, augmentation by reflection, rotation, shear and additions of salt noise. Previous approaches usually consisted an additional step of segmentation of the tumor region, a hard task for both analysis and computation. However, the statistical data retrieve from GLCM matrix do require such preprocessing in this project as well. The project is also novel in its design of CNN layer instead of using transfer learning. In case of transfer learning, about 140-170 layers of neural networks are used whereas for the present case about 16 layers seemed sufficient which yielded in some cases more accurate determination of tumor than previous approaches. Finally, implementation using only 16 layers considerably reduced the processing time from hours to about 10 mins for the same number of epoch and dataset.

5.2 Design Considerations

For our design, a tradeoff between speed and accuracy had to be made. In additions, for the general case, the MRI images are, in general, 3D images whose 2D versions were used for simplicity resulting in less accuracy. Since, available data are limited a rather arbitrary division of testing, training and validations had to be used. In designing the layers for CNN and choosing different hyperparameters trial and error had to be used since no concrete methods were found. We also had to limit ourselves in three different class of tumor due to a lack of dataset.

5.3 Investigations

5.3.1 Design of Experiment

Design of the CNN model was the primary challenge of the project. After its design, available dataset was partitioned as testing, training and validation as follows:

Table 1 Data Partitioning table

Purpose	Percentage
Training	68%
Validation	12%
Testing	20%

Following data partitioning, the network was trained with the training data and along its training validated by validation data. A validation patience of 10 was assumed with frequency of 25 iterations, chosen randomly. In additions, several other training options were chosen as tabulated:

Table 2 Training Options

Training Options	Values/Methods
Optimizers	Adam,SGD,RMSprop
Initial Learn Rate	64
Mini Batch size	64

Learn Rate Drop Factor	0.1
Validation Frequency	25
Maximum Epochs	20,50
Validation Patience	10

After the successful training, the trained network was saved in a .mat file and was used for testing by loading the .mat file.

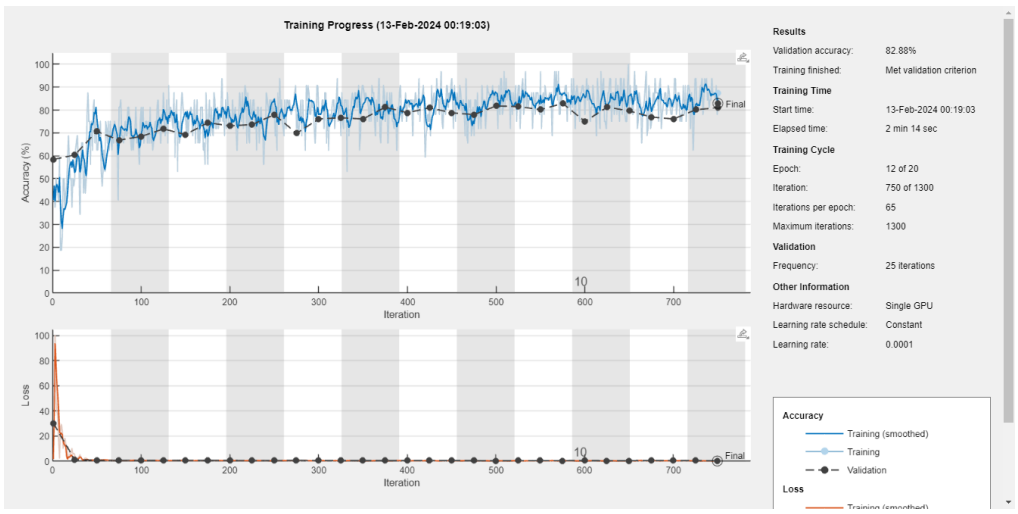


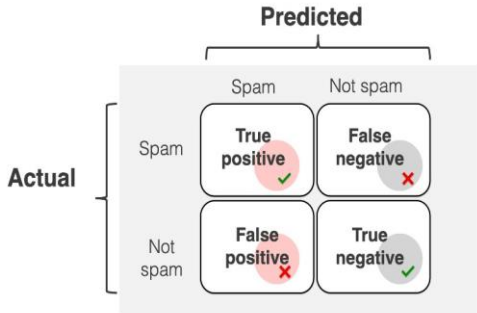
Figure 1 Training Progress

5.3.2 Data Collection

With the previously trained model loaded from .mat file, predictions for the test data were performed. These predicted tumor classes were compared to the original ones loaded as label names for the associated folder. The comparisons were presented in a confusion chart. In addition, accuracy and precision of the results obtained had to be performed. Accuracy and Precisions are defined as:

$$Accuracy = \frac{TP + TN}{P + N}$$

$$Precision = \frac{TP}{TP + FP}$$



A comparison of accuracy and precision for various classes and for the overall cases are collected for analysis purpose. Results for different optimizers were also obtained for comparison.

5.3.3 Result and Analysis

From the 20% test data run, the confusion matrix is obtained where horizontal axis represents predicted class and vertical axis represents true class. Now for the SGDM optimizer, the confusion matrix is:

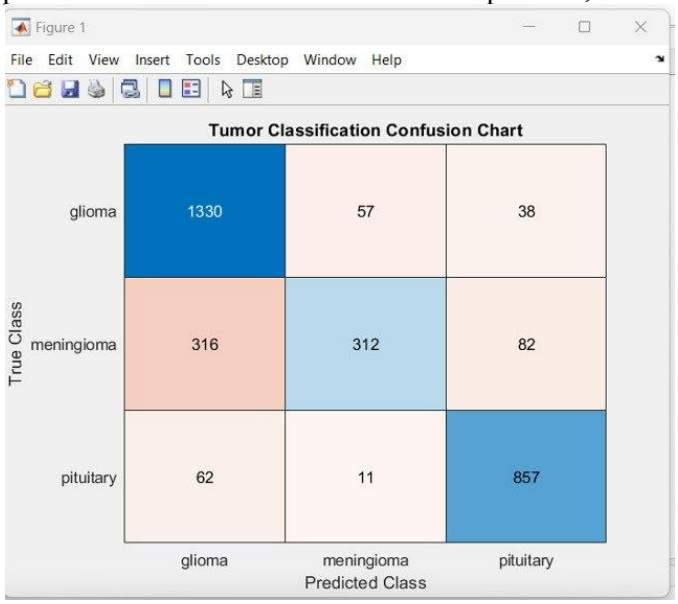


Figure 2 Confusion Matrix for SGDM

For the same optimizing algorithm, the accuracy and precision values are plotted below:

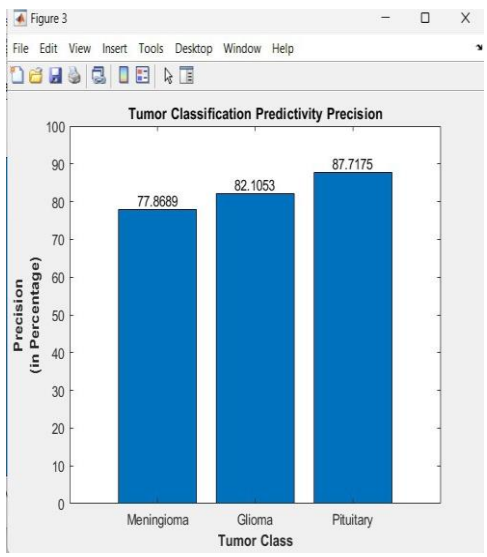


Figure 3 Precision values for SGDM

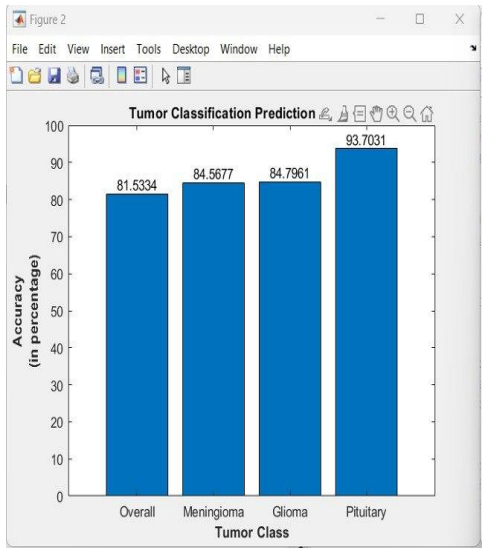


Figure 4 Accuracies for SGDM

Now, as the optimizer is changed, the confusion matrix changed for the case of RMSPROP and ADAM. For RMSPROP,

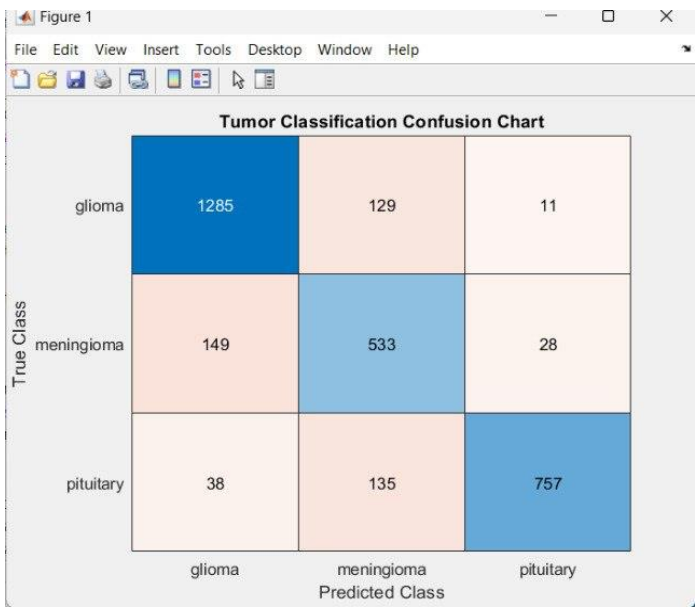


Figure 5 Confusion Matrix for RMSProp

Accuracy and precision values become,

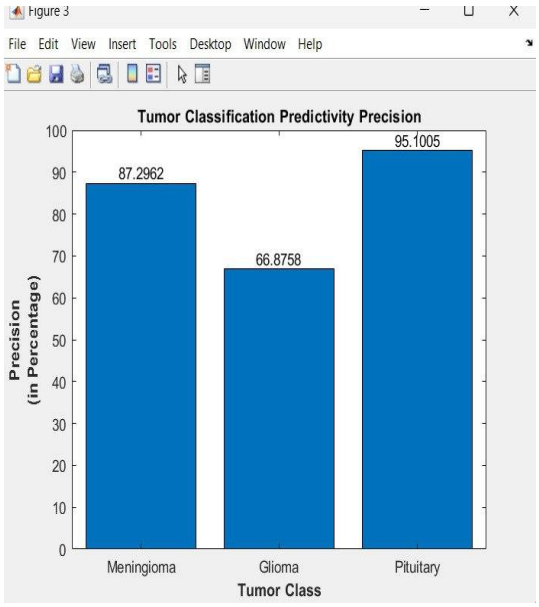


Figure 6 Precision values for RMSProp

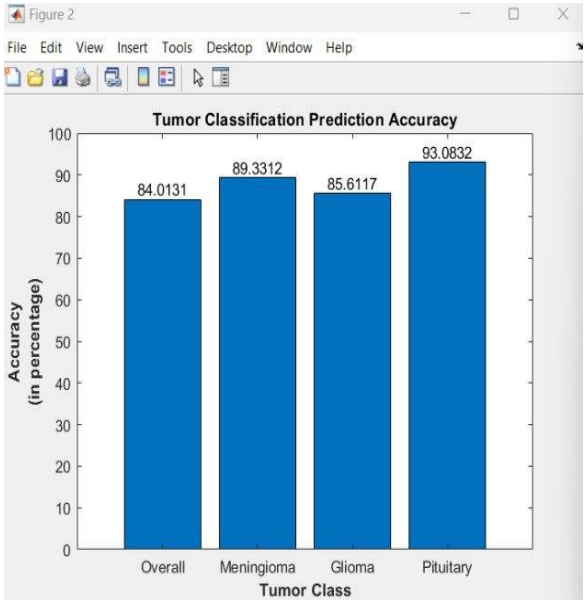


Figure 7 Accuracies for RMSprop

For adam, the results are,

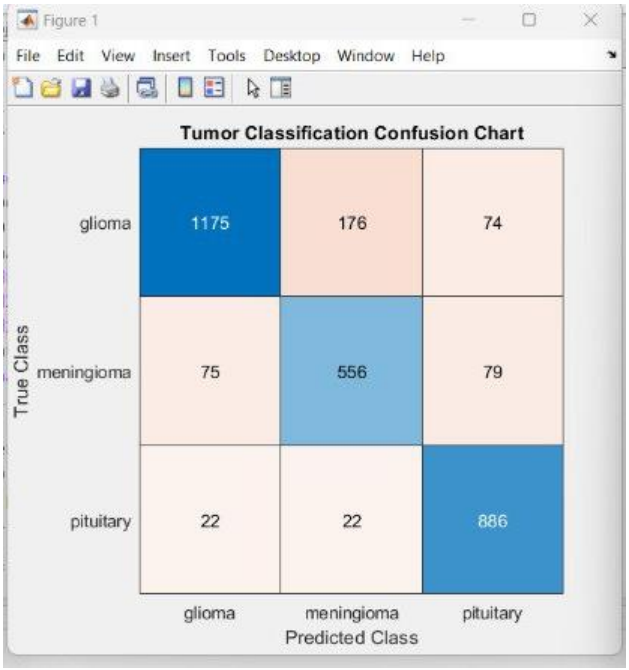


Figure 8 Confusion matrix for Adam

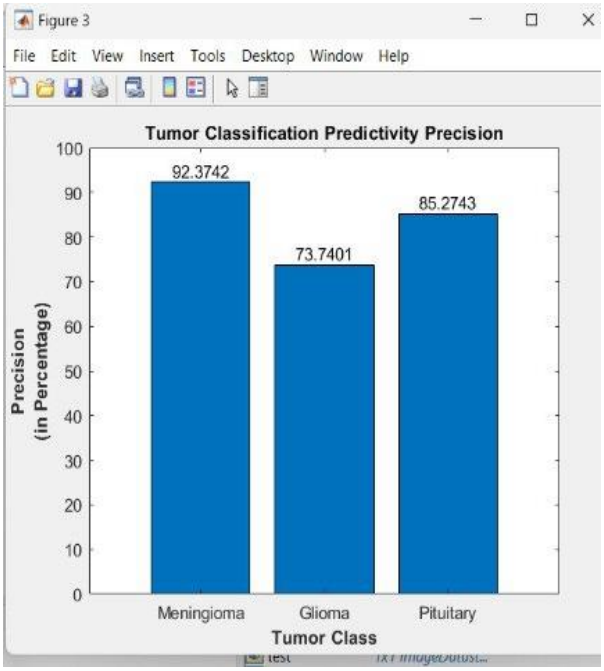


Figure 9 Precision values for Adam

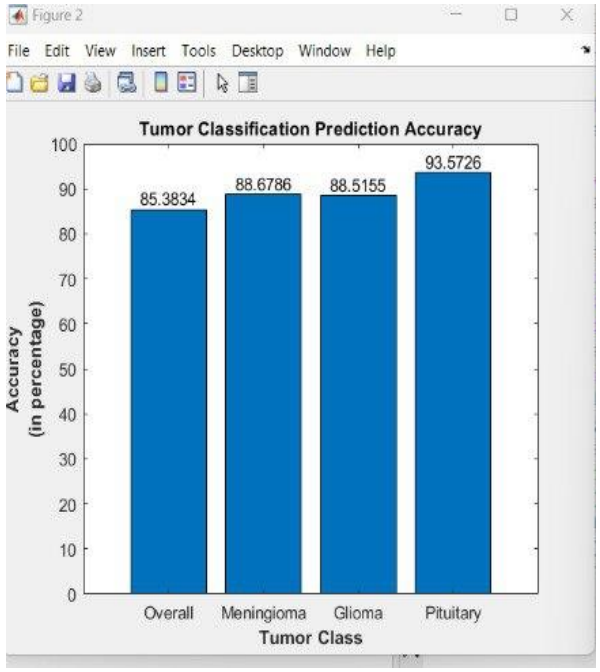


Figure 10 Accuracies for ADAM

5.3.4 Interpretations and Conclusions of Data

From the data thus obtained, it is apparent that ADAM provides the best of three optimizers with accuracy and precision increasing for all of the tumor classes. From confusion chart and precision histogram, higher error in designating meningioma and glioma can be observed with one mistaken for the other. This could be eliminated if the tumor region was segmented and its position in the skull in coordinated. Besides, Pituitary can be seen to be more accurately determined than the other with more than 95 percent of time. In conclusion, the present study provides a sufficiently accurate determination of the brain tumor which statement is supplemented by following table:

Table 3 Literature Review

Model	Best Accuracy	Classification Type	Classification Model
Cheng	91.21%	Binary	SVM and KNN
Paul	85.27%	Multi	CNN
Afshar	87.88%	Multi	CNN

Zacharaki	85%	Multi	SVM and KNN
Proposed Model	87.9%	Multi	CNN

5.4 Limitations of Tools

The present tool was implemented using MATLAB. The latest packages for deep learning are usually implemented in Python first. So, some of the tools are somewhat outdated in the present program. The project also implements only three types of tumor classification and does not indicate the stage or grade of tumor. The project also implemented some statistical properties for this purpose, nonetheless their applications were not implemented. Finally, the tumor region was identified which may have supplemented the program in tumor class detection by specialist.

5.5 Impact assessment

The following project demonstrates a program for detection of tumor classes categorized as Glioma, Meningioma, Pituitary. The project has an overall accuracy of about 90% which can be used in real medical services for the detection of tumor classes.

5.7 Ethical Issues

In the present project, the data used are personal and anonymous due to ethical reasons. Moreover, in the application of the program, ethical standards must be maintained. Since, the results obtained are in some sense approximate, they should be, ethically and ideally, supplemented by specialists' opinion on the subject.

6. Reflection on Individual and Teamwork (PO(i))

6.1 Individual Contribution of Each Member

Table 4 Individual Contributions

Different Segments of Project	Individual Contribution
Group Discussion	2006166, 2006172, 2006178, 2006190
Collecting Research Papers for the Brain Tumor Detection on the KNN Model	2006166, 2006190

Collecting Research Papers for the Brain Tumor Detection on CNN Model	2006172, 2006178
Design the Model of Our Project	2006166, 2006172, 2006178, 2006190
Design KNN model	2006172, 2006166
Design CNN model	2006178, 2006190
Implementation of the CNN model	2006166, 2006172, 2006178, 2006190
Analysis and Evaluation of Output	2006172, 2006190
Preparing Final Report	2006166, 2006172, 2006178, 2006190

6.2 Mode of the Teamwork:

Our team consisted of 4 members. Each team member brought a unique set of skills, perspectives, and experiences.

Project Planning: After discussing the project's scope, the many components that need to be completed gradually, its design, and its degree of difficulty with each other, the team decided on this project. Next, we made sure that each project component was properly planned.

Task Distribution: Following a successful project planning phase, we divided up the various project tasks among the team members.

Design Project Model: After finalizing the project we explore many resources. Then we design a model of our project. Then we work according to the flow chart of that model and implement our project.

Software Implementation: software implementation is the key part of this project. First, we had to implement image processing and the main challenge of our project was to design CNN model. We implemented our project by using Matlab.

Communication and Discussion: We held multiple group talks during the project. Everyone in

our group was free to maintain their opinions. All of our decisions were made with the team members' approval and following a meeting.

6.3 Diversity Statement of Team:

Every one of the team members has their skills and expertise. We combined all the skills and expertise to complete our project. Whenever a problem arose, we were able to solve it because of the diverse knowledge of the team.

6.4 Log Book of Project Implementation:

Table 5 Log Book of Project Implementation

Date	Achievement	Individual Role	Team Role	Comments
20.01.24 & 21.01.24	Group Discussion	2006166, 2006172, 2006178, 2006190 joined the group discussion	We focused on the theoretical background of the project and completed project planning	
26.01.24	Collecting Research Papers for the Brain Tumor Detection on the KNN Model	2006166, 2006190 Collected the research papers on the weighted least square method		
27.01.24	Collecting Research Papers for the Brain Tumor Detection on CNN Model	2006172, 2006178 Collected the research papers on the weighted least square method		
10.02.24	Design the Model of Our Project	2006166, 2006172, 2006178, 2006190	All members were present so that we could solve any problem	

			we face easily	
27.02.24	Design KNN model	2006172, 2006166		
27.02.24	Design CNN model	2006178, 2006190		This part was critical in our project
27.02.24	Implementation of the CNN model	2006166, 2006172, 2006178, 2006190		
28.02.24	Analysis and Evaluation of Output	2006172, 2006190		
28.02.24 & 29.02.24	Preparing Final Report	2006172 divided the report writing tasks for all members and compiled it after collecting different parts from different member	Other member completed their respective portion of the report within the given period	

7 Communication to External Stakeholders (PO(j))

7.1 Executive Summary

Our project focused on detecting brain tumors using MRI images, employing a deep-learning CNN model. We designed our project based on research papers and implemented it using Matlab. With a team of four members, each contributed to different aspects of the project. One member handled image augmentation, while another focused on training and testing. Additionally, one member was responsible for the design of the CNN model. To properly implement the project in software, we integrated individual contributions in Matlab to create the final project. For analysis and evaluation of the output, all team members assessed accuracy,

precision, and the confusion chart, yielding satisfactory results. Furthermore, our computational time was lower than others, which added to our satisfaction with the project.

8 Future Work (PO(l))

In addition to the current scope, our project lays the foundation for future advancements in brain tumor detection and classification. Firstly, we aim to expand our model to encompass more tumor classes beyond the three currently included. This expansion will enhance the utility of our model for researchers and doctors involved in tumor classification, providing a comprehensive tool for accurate diagnosis across various tumor types. Furthermore, we plan to integrate a CNN model for tumor region segmentation, allowing for precise localization and delineation of tumor boundaries within MRI images. By incorporating these future works, our project strives to further contribute to the field of medical imaging and improve outcomes for patients with brain tumors.

9 References

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